



# Flood Polynomial Model — Test Results

Automated Test Documentation for Model Governance

MKM Research Labs

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# 1 Test Results — Flood Polynomial Model (MKM-FPO-001)

Tests run on **04 April 2026 at 18:43:46**.

Table 1: Test Results Summary — Flood Polynomial Model (MKM-FPO-001)

|             |       |
|-------------|-------|
| Total Tests | 33    |
| Passed      | 33    |
| Failed      | 0     |
| Skipped     | 0     |
| Duration    | 0.00s |

Table 2: Detailed Test Results — Flood Polynomial Model

| Test Description                                                      | Acceptance Criteria                                                                    | Result | Time   |
|-----------------------------------------------------------------------|----------------------------------------------------------------------------------------|--------|--------|
| Coefficients used by model match config/models.py.                    | hasattr(flood_poly_coeffs, 'a');<br>hasattr(flood_poly_coeffs, 'f') (+1 more)          | Pass   | 0.000s |
| Fit metadata from config                                              | flood_poly_fit["model_id"]<br>== "flood-poly-v1";<br>flood_poly_fit["r_squared"] > 0.9 | Pass   | 0.000s |
| Log eps from config                                                   | flood_poly_log_eps == 1e-8                                                             | Pass   | 0.000s |
| Storm hours from config                                               | flood_poly_storm_hours == 168                                                          | Pass   | 0.000s |
| Coefficients count                                                    | card["num_parameters"] == 6                                                            | Pass   | 0.000s |
| Fit quality has metrics                                               | "r_squared" in fq; "mae" in fq (+1 more)                                               | Pass   | 0.000s |
| Has required fields                                                   | "model_id" in card; "version" in card<br>(+3 more)                                     | Pass   | 0.000s |
| Model id matches config                                               | card["model_id"] ==<br>flood_poly_fit["model_id"]                                      | Pass   | 0.000s |
| Returns dict                                                          | isinstance(card, dict)                                                                 | Pass   | 0.000s |
| Water far above severe at late hour should be near 1.0.               | p > 0.9                                                                                | Pass   | 0.000s |
| Water well above severe threshold should give high P(flood).          | p > 0.8                                                                                | Pass   | 0.000s |
| hour=167 (last storm hour) should not raise.                          | 0.0 <= p <= 1.0                                                                        | Pass   | 0.000s |
| hour=0 should not raise.                                              | 0.0 <= p <= 1.0                                                                        | Pass   | 0.000s |
| P(flood) at hour 0 should be lower than mid-storm for moderate water. | p_early < p_mid                                                                        | Pass   | 0.001s |
| In range                                                              | 0.0 <= result <= 1.0; 0.0 <= result<br><= 1.0                                          | Pass   | 0.000s |

| Test Description                                             | Acceptance Criteria                                                                                                          | Result | Time   |
|--------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Water well below severe threshold should give low P(flood).  | <code>p &lt; 0.1</code>                                                                                                      | Pass   | 0.000s |
| Higher water level should give higher P(flood) at same hour. | <code>p_low &lt; p_mid &lt; p_high</code>                                                                                    | Pass   | 0.000s |
| Negative severe returns zero                                 | <code>p_flood.poly(water_level=5.0, hour=48, severe_level=-1.0) ...;</code><br><code>p_flood.simple(5.0, -1.0) == 0.0</code> | Pass   | 0.000s |
| Returns float                                                | <code>isinstance(result, float);</code><br><code>isinstance(result, float)</code>                                            | Pass   | 0.000s |
| Near-zero water level should give near-zero P(flood).        | <code>p &lt; 0.01</code>                                                                                                     | Pass   | 0.000s |
| Zero severe returns zero                                     | <code>p_flood.poly(water_level=5.0, hour=48, severe_level=0.0) ...;</code><br><code>p_flood.simple(5.0, 0.0) == 0.0</code>   | Pass   | 0.000s |
| All in range                                                 | <code>all(0.0 &lt;= p &lt;= 1.0 for p in result)</code>                                                                      | Pass   | 0.000s |
| Correct length                                               | <code>len(result) == 5</code>                                                                                                | Pass   | 0.000s |
| Empty list                                                   | <code>p_flood.series([], severe_level=6.58)</code><br><code>== []</code>                                                     | Pass   | 0.000s |
| Returns list                                                 | <code>isinstance(result, list)</code>                                                                                        | Pass   | 0.000s |
| Above severe capped at one                                   | <code>p_flood.simple(10.0, 6.58) == 1.0</code>                                                                               | Pass   | 0.000s |
| At severe equals one                                         | <code>p_flood.simple(6.58, 6.58) == 1.0</code>                                                                               | Pass   | 0.000s |
| In range                                                     | <code>0.0 &lt;= result &lt;= 1.0; 0.0 &lt;= result &lt;= 1.0</code>                                                          | Pass   | 0.000s |
| Negative severe returns zero                                 | <code>p_flood.poly(water_level=5.0, hour=48, severe_level=-1.0) ...;</code><br><code>p_flood.simple(5.0, -1.0) == 0.0</code> | Pass   | 0.000s |
| $P = \min(1, w/s)$ .                                         | <code>abs(p_flood.simple(3.29, 6.58) - 0.5) &lt; 0.01</code>                                                                 | Pass   | 0.000s |
| Returns float                                                | <code>isinstance(result, float);</code><br><code>isinstance(result, float)</code>                                            | Pass   | 0.000s |
| Zero severe returns zero                                     | <code>p_flood.poly(water_level=5.0, hour=48, severe_level=0.0) ...;</code><br><code>p_flood.simple(5.0, 0.0) == 0.0</code>   | Pass   | 0.000s |
| Zero water                                                   | <code>p_flood.simple(0.0, 6.58) == 0.0</code>                                                                                | Pass   | 0.000s |

## 1.1 Test Document History

| Date        | Version | Description                   | Author   |
|-------------|---------|-------------------------------|----------|
| 04-Apr-2026 | 1.0     | Initial test suite (33 tests) | D. Kelly |